# Page 1: General Bearing Overview

Standard operating procedures for industrial machine bearings. Regular inspection prevents catastrophic motor and axle failure. Check for unusual noise, heat accumulation, and excessive vibration during daily rounds.

# Page 2: Bearing Wear Diagnosis & Action

Defect Type: Bearing Wear and Fatigue Scoring.

Symptom: Circular scoring marks, metallic noise, or overheating in rotational assembly.

Action Required: Immediately isolate machine power. Remove bearing housing and replace worn ball bearings with part #BRG-9920. Apply high-temperature synthetic grease prior to restarting operations. Priority: High.

# Page 3: Bearing Overheating and Lubrication Failure

Defect Type: Bearing Overheating.

Symptom: Housing temperature above 95 C, discoloured (blue or brown) races, burnt grease smell, grease leaking past the seals.

Action Required: Stop the machine and let it cool for 30 minutes. Purge old grease, inspect the seals and re-lubricate with grease type HT-2. If races are discoloured, replace the bearing set. Do not exceed 30 percent grease fill. Priority: High.

# Page 4: Bearing Installation, Fit and Torque

Procedure: Installing a replacement bearing.

Heat the inner ring with an induction heater to between 80 and 100 C; never use an open flame. Slide the bearing onto the shaft without hammering. Tighten the lock nut to 85 Nm and bend the lock washer tab. Check shaft runout is below 0.02 mm with a dial gauge. Priority: Medium.

## Page 5: Spare Parts and Maintenance Records

Keep two spare bearing sets (part #BRG-9920) in the stores. Record every bearing change in the maintenance log with date, running hours, operator and the failure mode found. Review the log monthly to find repeat failures.